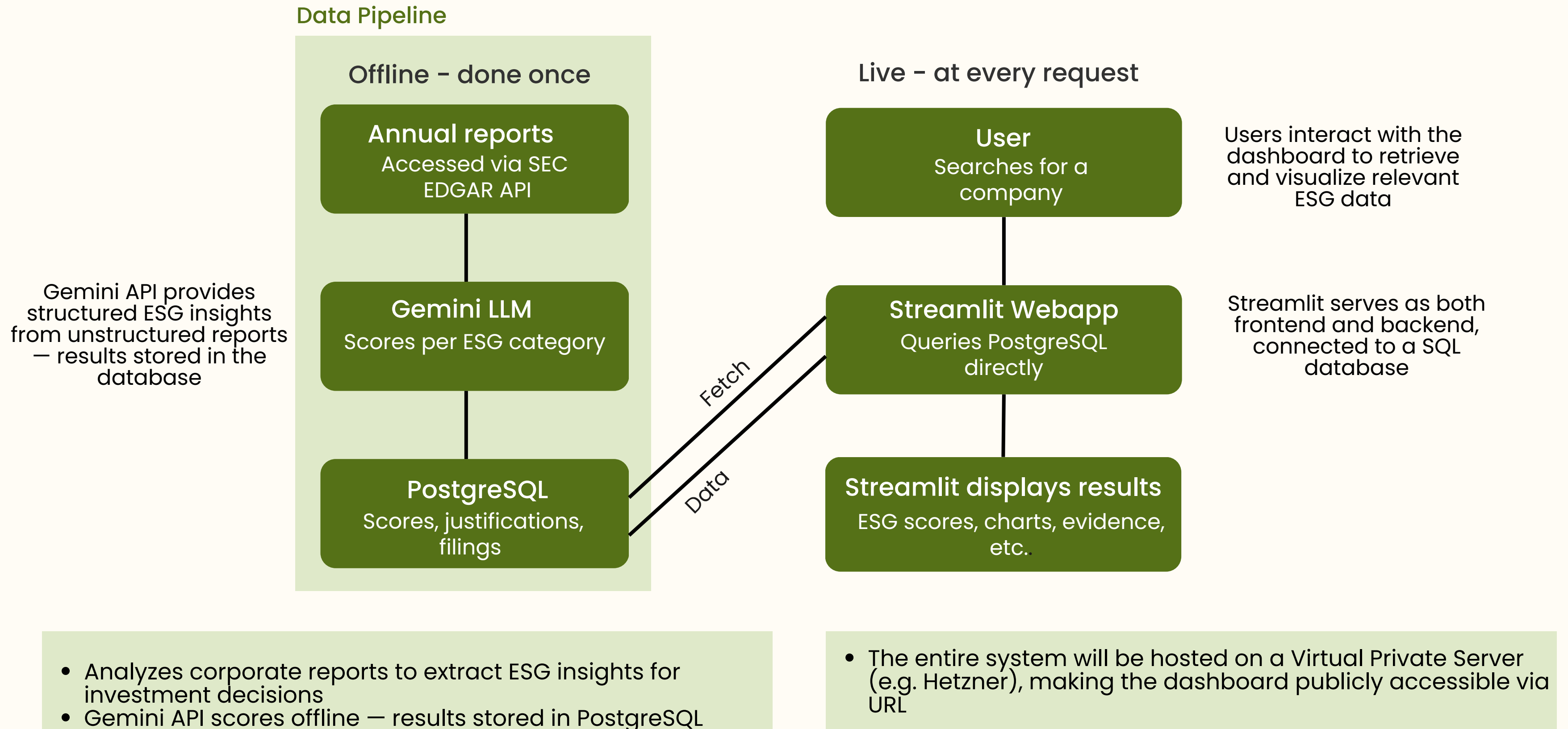
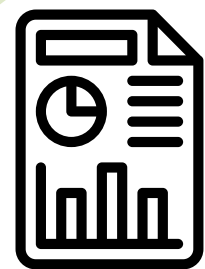


System Architecture

ESG Analytics of corporate reports

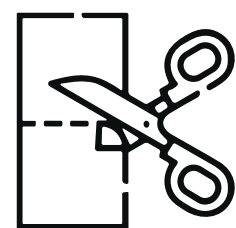


Data Pipeline



Financial Report Retriever

- Queries SEC EDGAR public API
- Download financial data
- Focus on financial sector
- Using Python requests library
- Parses HTML via BeautifulSoup



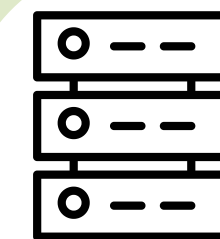
Text Chunking

- Splits each report into segments and produces chunks
- Keeps only ESG-relevant passages
- Reduces number of LLM API calls



LLM Scoring

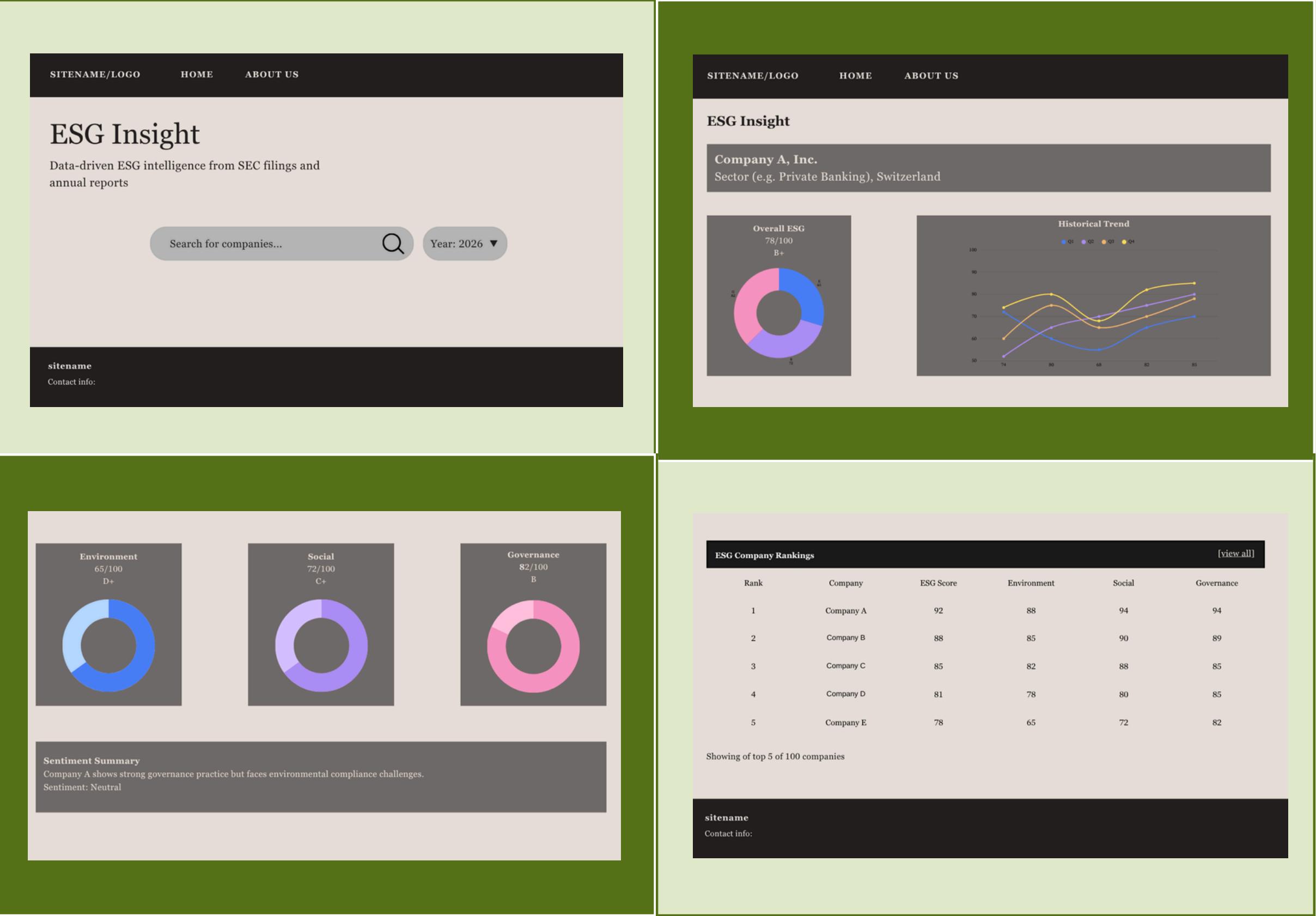
- Calls Google Gemini API
- Sends ESG-relevant chunks as context
- Assigns score per sub-category and justification
- Runs offline – scores are stored, not live



SQL Database

- PostgreSQL database with 4 relational tables
- Allows querying and ranking companies by ESG dimension
- Serves as the single source of truth

Dashboard



Interface Layer

- Company search
- Year selector
- ESG overall score chart
- E/S/G score individually
- Historical trend line
- Company ranking table

Data Integration

- Displays LLM-generated scores (Gemini API)
- Queries company rankings